

Which Test Answers the Question Asked?

Unit 3 · Topic 3.5 · THE PROBLEM

A district of 2,400 students had bus ridership $p_0 = 0.40$ before a new route. Before collecting data, the director asks whether the current proportion p has *increased*. An SRS of 180 current students finds 84 riders.

Current SRS counts

Group	Bus	Other	Total
Students	84	96	180

Maya: “Use $H_0 : p = 0.40$ versus $H_a : p > 0.40$ and a one-sample z -test. Check large counts with 72 and 108.”

Theo: “Use $H_a : p \neq 0.40$ because samples can differ either way. Check large counts with the observed 84 and 96.”

FIRST TAKE — one sentence before you work the parts

Does the director want to detect any change, or only more bus riders? Use a phrase from the study.

- DOK 1** (a) Define the population parameter p in context and identify the null value p_0 .
- DOK 2** (b) Compute $\hat{p}$, name the testing procedure, and verify the random, 10 percent, and large-counts conditions using the appropriate values.
- DOK 3** (c) In 3–4 sentences, choose a setup using the director’s word increased. Explain why the count check uses 72 and 108 rather than 84 and 96. State what different question Theo’s alternative would answer.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.